ModelIt!

Teacher's Guide

The Secret War in Your Backyard

How Ecosystems Balance Competition, Cooperation, and Survival

Quick Reference

Grade Level	Grade 6
NGSS Standard	MS-LS2-2, MS-LS2-3, MS-LS2-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-LS2-2, MS-LS2-3, MS-LS2-5

Construct an explanation that predicts patterns of interactions among organisms across multiple ecosystems. Develop a model to describe the cycling of matter and flow of energy among living and nonliving parts of an ecosystem. Evaluate competing design solutions for maintaining biodiversity and ecosystem services.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how predator and prey populations interact with habitat size and resource availability to create ecosystem dynamics.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems / LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Organisms and populations of organisms are dependent on their environmental interactions both with other living things and with nonliving factors. Ecosystems are dynamic and can respond to disruptions.
Crosscutting Concept	**Stability and Change** — Students explore how ecosystems maintain dynamic stability through population cycles, and how disruptions (habitat loss, species removal) can push systems past tipping points into instability.

CAST Testing Alignment

MULTIPLE CHOICE:

In a grassland ecosystem, a disease kills 80% of the rabbit population. Which of the following is the most likely short-term effect on other species in the ecosystem?

CONSTRUCTED RESPONSE:

A state park manager is considering reintroducing wolves to control an overpopulated deer herd. Using ecosystem concepts, predict what would happen to the deer population, vegetation, and overall biodiversity if wolves were reintroduced.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Ecology is the study of how organisms interact with each other and their environment. Every ecosystem — from your backyard to the Amazon — operates on the same fundamental principles of competition, predation, cooperation, and energy flow.

Predator-Prey Dynamics

Predator and prey populations naturally cycle in a predictable pattern: when prey is abundant, predators thrive and increase. More predators reduce prey numbers. Fewer prey causes predators to decline. Fewer predators allows prey to recover. This Lotka-Volterra cycle repeats indefinitely in a healthy ecosystem. The famous example is the lynx and snowshoe hare cycle in Canada, documented over 100+ years.

Trophic Cascades

When a top predator is removed, the effects cascade through the entire food web. In Yellowstone, removing wolves caused elk to overgraze riverbanks, which caused erosion, which degraded streams, which reduced fish populations, which affected bears and eagles. When wolves were reintroduced in 1995, the entire ecosystem recovered — even the rivers changed course. This is called a trophic cascade.

Carrying Capacity

Every environment has a maximum population it can support — its carrying capacity. This limit is determined by available food, water, space, and shelter. When a population exceeds carrying capacity, individuals compete for scarce resources, death rates increase, and the population crashes. Habitat loss reduces carrying capacity, making population crashes more likely.

Biodiversity and Resilience

Ecosystems with more species are more resilient to disturbance. If one species is lost in a diverse ecosystem, others can fill its ecological role. In a simple ecosystem with few species, losing one can collapse the whole system. This is why biodiversity conservation is critical — it's not just about saving individual species, it's about maintaining ecosystem function for all life, including humans.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — **Locate the System**
Students identify predator population, prey population, habitat size, and resource availability as the key components driving ecosystem dynamics.

E — **Establish Relationships**
Students discover the cyclical relationships: prey positively affects predators (more food), predators negatively affect prey (consumption), and both depend positively on habitat and resources.

V — **Visualize & Model**
Students build a model showing how predator-prey cycles emerge from simple component interactions and how external changes disrupt these cycles.

E — **Evaluate Outcomes**
Students run balanced, predator-removal, and habitat-loss scenarios to observe how ecosystems respond to different types of disruption.

R — **Revise & Extend**
Students add invasive species or decomposer activity to explore more complex ecosystem interactions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Right now, in every yard and park and forest, there's a war happening that you can't see."*

Watch For

Show a time-lapse video of a backyard ecosystem or a nature documentary clip.

Slide 2: Learning Objectives

3 min

Say: *"Today we're going to model the invisible battles that keep ecosystems alive."*

Watch For

Have students read objectives. Pre-teach 'biodiversity,' 'carrying capacity,' and 'trophic level.'

Slide 3: Big Question

3 min

Say: *"What would happen if all the predators in your backyard disappeared? It seems like prey animals would be happy — but would they?"*

Watch For

Quick write: Students predict what happens if all predators are removed. Most will be wrong!

Slide 4: LEVER Framework

2 min

Say: *"We'll model a real ecosystem and then BREAK it to see what happens."*

Watch For

Preview LEVER steps. Tell students they'll be ecosystem managers making life-or-death decisions.

Slide 5: Activity 1: Components

8 min

Say: *"What factors determine whether an ecosystem stays balanced or collapses?"*

Watch For

Guide sorting. Discuss why habitat size and resources are external (environment), while populations are internal (they change in response).

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the mind-bending part: more predators is actually GOOD for the ecosystem, not bad."*

Watch For

Draw the cycle on the board: prey up → predators up → prey down → predators down → repeat.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's see what happens when we remove all predators. Paradise for prey — or disaster?"*

Watch For

Students predict, then run all three scenarios. The predator-removal result is counterintuitive and powerful.

Slide 8: Discoveries

5 min

Say: *"Predators don't just kill prey — they SAVE ecosystems. That's why losing wolves was a catastrophe for Yellowstone."*

Watch For

Share the Yellowstone wolf reintroduction story. Connect to model results.

Slide 9: STEM Challenge

12 min

Say: *"A nature preserve is collapsing. You're the ecologists. Use your model to design a rescue plan."*

Watch For

Distribute ecosystem data sheets. Groups design restoration plans with model evidence.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Habitat Size, Resource Availability
- **INTERNAL:** Predator Population, Prey Population

Relationships

- **Prey Population to Predator Population:** POSITIVE (+) — More prey means more food available for predators, allowing predator populations to grow. When prey is abundant, predators reproduce more and survive longer.
- **Predator Population to Prey Population:** NEGATIVE (-) — More predators consume more prey, reducing the prey population. This creates the classic predator-prey cycle where both populations oscillate.
- **Habitat Size to Carrying Capacity:** POSITIVE (+) — Larger habitat provides more resources, shelter, and territory, increasing the maximum population the environment can support for both predators and prey.

Key Discoveries

- Predator and prey populations naturally cycle — when prey increases, predators follow, then prey crashes, then predators crash
- Removing predators causes prey to overshoot carrying capacity, depleting resources and causing a population crash
- Habitat loss reduces carrying capacity for ALL species, often pushing the smallest populations to extinction first
- Biodiversity makes ecosystems more resilient — diverse ecosystems recover from disturbances faster than simple ones

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Predators are bad for ecosystems"	Predators are ESSENTIAL for ecosystem health. They prevent prey from overshooting carrying capacity, keep ecosystems balanced, and maintain biodiversity. Removing top predators consistently leads to ecosystem degradation. The Yellowstone wolf story is the most famous demonstration of this principle.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Ecosystems will balance themselves no matter what"	While ecosystems are resilient to small disturbances, they have tipping points beyond which they cannot recover. If habitat loss, pollution, or species removal pushes an ecosystem past its tipping point, it can collapse to a fundamentally different state that may take centuries to recover — if it can recover at all.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Only large animals matter in ecosystems"	Every species plays a role, and some of the most important are the smallest. Insects pollinate plants, decomposers recycle nutrients, bacteria fix nitrogen, and plankton produce half the world's oxygen. Losing a 'small' species can have cascading effects just as dramatic as losing a large predator.